

Active Beauty Scientific File



Endothelyol™

Scientific File Endothelyol™
Version: 02 -- Date: 07/2018

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1. Summary of the scientific results

IN VITRO (on mastocytes)

- Mast cells treated with ENDOTHELYOL™ at 2.5 %, 1.5 % and 1% showed an inhibition of the spontaneous histamine release, even in neuropeptidergic stimulated conditions.

IN VITRO (on human normal keratinocytes extracted from skin biopsies)

- ENDOTHELYOL™ **inhibits IL8 release** upon IL1 β cell stimulation or not at 2.5 % (58 % inhibition and 67 %, respectively for not stimulated and stimulated cells) and at 1 % (45 % inhibition and 43 %, respectively for not stimulated or stimulated cells. These results are very significant, $p < 0.01$).
- ENDOTHELYOL™ **inhibits TNF α release** at 2.5 %, 1.5 % and 1%. (% inhibition 37%, 39 and 30%, respectively $p < 0.01$). Dexametazone used as positive control inhibits the production of this mediator by 59 % at 10 μ M.
- The treatment of human normal keratinocytes with ENDOTHELYOL™ at 1 % ; 0.25% and 0.1% **inhibits the product on of PGE₂** induced by PMA. This result is statistically significant (% inhibition 63 %, 63 % and 64 %, respectively).
- The treatment of human normal keratinocytes with ENDOTHELYOL™ at 2.5 % and 1.5 % **inhibits the production of VEGF** induced by IL1 β (inhibition 37%, $p < 0.05$ and 33 %, $p < 0.01$, respectively)

IN VITRO (on human endothelial cells cultured in 3D)

– ENDOTHELYOL™ shows a significant inhibitory effect on the development of neoPvessels at 1,5% and 2,5%.

IN VIVO (clinical evaluation on a panel of volunteers versus placebo)

- Dermatologist: the serum was tested under dermatological control and no adverse cutaneous reaction was observed. The dermatologist concluded to a very good tolerance of the products tested.
- Results on rosacea: after 28 days of application, a 12% improvement of rosacea is observed on the cheekbones.
- Results on melanin: after 28 days of application, a 8% decrease of melanin is observed on the cheekbones.

2. In vitro assay on histamine release

Effect of LBG 10-602 on histamine release induced by substance P in rat peritoneal mast cells

2.1. Introduction

Mast cells play an important role in the pathophysiological changes observed in cutaneous allergic and inflammatory reactions. In the skin, they are located in the vicinity of the dermo-epidermic junction. Serosal mast cells present the same functional characteristics as human cutaneous mast cells, so they represent a good experimental model of human mast cells. They release preformed



mediators such as histamine and hydrolytic enzymes after the activation with calcium Ionophore (A 23187). Furthermore, recent studies indicate that the nervous system may also influence cutaneous inflammatory events through the activation of immune cells by substance P, a neurotransmitter. Substance P is released in the skin. It can bind to and activate mast cells of the serosal type resulting in histamine and tryptase release.

The aim of the present study was to investigate the effect of LBG 10-602 on substance P-induced histamine release in rat peritoneal mast cells.

2.2. Materials and methods

2.2.1. Reagents

Preparation of substance P

Substance P is a neuropeptide, purchased from NeoMPS (ref. SC000, Strasbourg). The product is dissolved at 1 mM and used at 50 μ M to induce the release of tryptase.

Preparation of the products to be tested

LBG 10-602 is prepared from a sterile stock solution and tested at 2.5% – 1.5% and 0.25% (v/v, dilutions in medium culture).

A calcium solution is used as inhibitory control.

2.2.2. Isolation of rat peritoneal mast cells

Mast cells from male Wistar rats (200-300 g; Charles River, France) were obtained following an intraperitoneal injection of 15 ml of Tyrode's buffer (KCl 3.5 mM ; NaCl 140mM ; Glucose 5.5 mM ; Hepes 10 mM, pH 7.2).

After injection, the peritoneal cavity was massed for 2-3 min and the fluid collected. The cell suspension was centrifuged (1500 rpm, 4°C) and the pellet was resuspended in 1 ml of Tyrode buffer. Mast cells were stained using toluidine blue solution and counted. The number of mast cells per rat varied from $0.5 \cdot 10^6$ at $2 \cdot 10^6$.

2.2.3. Treatment and stimulation of mast cells

Cells ($1 \cdot 10^5$ /ml or 50000 cells/tube) were incubated at 37°C in the presence or not of the products during 30 min and then stimulated substance P during 15 min. The stimulations were stopped by centrifugation at 4°C.

The cytotoxicity of the products were assessed by measurement of the Lactate Deshydrogenase (assay LDH, Roche Diagnostics, France), in the supernatants from LBG10-602-treated cells after 45 min of incubation. The LDH activity is determined in an enzymatic test: in the first step, NAD⁺ is reduced to NADH/H⁺ by the LDH- catalyzed conversion of lactate to pyruvate. In the second step, the catalyst transfers H/H⁺ from NADH/H⁺ to the tetrazolium salt which is reduced to formazan. An increase in the amount of dead or damaged-cells results in an increase of LDH enzyme activity in the culture supernatant. Each concentration is evaluated in duplicate and each assay is measured in duplicate. We obtained 4 values for absorbance reading at 490 nM.



2.2.4. Histamine release

The reaction is stopped by centrifugation (1800 rpm, 10 min. at 4°C). Supernatants and cell pellets are suspended with perchloric acid 0,4N. The measurement is performed with an ELISA kit Histamine (IBL Hamburg, RE59221). Each tube is measured in duplicate.

2.3. Results

2.3.1. LDH release

In this experiment, we observe an interference with the product LBG 10-602, so that the obtained results could not be interpreted.

Blanc products (in LDH test)	0%	2.5%	1.5%	1%	0.25%
Absorbance 490 nm	0.0062	2.3859	2.3045	0.819	0.3881

Cell viability of LBG10-602 was verified with blue Trypan's exclusion test by viable cells.

LBG-10602 is not cytotoxic at 2.5% - 1.5% - 1% and 0.25%.

2.3.2. Histamine release

For each tested condition, the results present the means of 3 values « not stimulated » and 3 values « stimulated » by substance P, measured in duplicate.

The percentage of inhibition is expressed as follow: $1 - \left(\frac{\text{histamine release from treated cells}}{\text{control}} \right) \times 100$

* Student's t test - comparison of means, n=3 values – significant at 95% if $p < 0.05$ * and at 99% if $p < 0.01$ **.

	Not stimulated	% Inhibition	Substance P	% Inhibition
Control	39.3 ± 2.3		131.7 ± 11.5	
LBG10-602 2.5%	31.7 ± 1.5	19% **	111.7 ± 7.6	15% *
LBG10-602 1.5%	31 ± 1.7	21% **	103.7 ± 7.1	21% *
LBG10-602 1%	32 ± 0.0	19% **	106.7 ± 2.9	19% *
LBG10-602 0.25%	36.3 ± 1.2	8% $p=0.0697$	122 ± 24.6	7% ns
Calcium 10 mM			45.3 ± 2.3	66% **



2.4. Conclusion

Mast cells treated with LBG 10602 at 2.5 %, 1.5 % and 1% induced an inhibition of the spontaneous histamine release or produced by a neuropeptidergic stimulation.

In conclusion, these results suggest that LBG 10-602 is a good candidate to inhibit the neurogenic inflammatory cutaneous reactions and associated pain.

3. *In vitro* assay of anti-inflammatory properties: measure TNF- α and Interleukine 8

Effect of LBG10-602 on IL-8 and TNF- α release by normal human keratinocytes

3.1. Introduction

Inflammation is a normal physiologic and immediate response to any infectious or mechanical aggression. Keratinocytes are involved in the immune defense: they secrete cytokines among which Interleukine-8 (IL-8) which induces a chemotactic gradient attracting cells implicated in inflammation, particularly neutrophils. These cells infiltrate irritated areas and release proteolytic enzymes like elastase. This enzyme stimulates IL8 release in the skin. Furthermore, an increase of its activity was associated with a degradation of matrix skin proteins, like elastin, collagen or proteoglycans resulting in skin aging and reducing skin elasticity.

Normal human keratinocytes (KHN) release interleukin 8 (IL8) under stimulation with Interleukine-1 β . They also produce TNF- α upon stimulation with Phorbol Myristate Acetate (PMA).

The aim of this study is to evaluate the effects of NHK treatments with LBG 10-602 on IL8 and TNF- α release upon suitable cells stimulations.

3.2. Products cytotoxicity – XTT assay

The aim of this study is to evaluate the cytotoxicity of LBG 10-602 on normal human keratinocytes (primocultures).

XTT assay (non radioactive and colorimetric method allows to quantify viability and proliferation of cells through their mitochondrial dehydrogenase activity. This is an easy and accurate method, with reproducible results.

3.2.1. Assay principle

The assay is based on the cleavage of the yellow tetrazolium salt XTT to form an orange formazan dye by metabolic active cells. Therefore, this conversion only occurs in viable cells. The formazan dye formed is soluble in aqueous solutions and is directly quantified using a scanning multiwell spectrophotometer (ELISA reader, wavelength 450nm).

3.2.2. Reagents

“Cell Proliferation Kit II (XTT)” (Roche) contains XTT labeling reagent and PMS (N-methyl dibenzopyrazine methyl sulfate, electron-coupling reagent). The mixture is prepared freshly at 0.3 mg/ml final concentration in culture medium KSFM (Gibco Invitrogen).



3.2.3. Product to be tested

LBG10-602 was supplied in liquid form: 2 bottles of 2 ml sterile stock solution to be diluted in culture medium. Keeping at 4°C.

Tested concentrations: 2.5 - 2 - 1.5 - 1 0.5% and 0.25 % (v/v, dilutions in medium culture)

3.2.4. Cell culture

Human normal keratinocytes (HNK, isolated from biopsy obtained from surgical operation) are plated in 96-well microplate at the density of 20 000 cells/well (culture medium KSFM – Gibco® Invitrogen). Plate is placed in an incubator maintained at 37°C and 5% CO₂ until cells are confluent at 80%.

3.2.5. Cell treatment

Cells are treated at different concentrations in culture medium for 48h. Each product concentration is evaluated on 4 wells. Cells are kept without treatment for control condition.

3.2.6. Cytotoxicity assay

After 48h incubation period, cellular wells are rinsed with PBS without Ca⁺⁺/Mg⁺⁺ (Gibco® Invitrogen). XTT mixture is added to each well and the plate incubated for 3h. The spectrophotometrical absorbance is measured using a microtiter plate (Safire, Tecan). The wavelength to measure absorbance of the formazan is 450 nm.

Absorbance measurement from 4 wells treated with the same product concentration are averaged. This average is compared with control average (4 wells also): Student's t test – significant difference at 95% if p<0.05* and at 99% if p<0.01**.

Viability from cells treated with the products are defined as percentage compared with control 100% (non treated cells) :

$$\% = [(OD_{\text{Treatment}} / OD_{\text{control}})] \times 100$$

3.3. Quantification of IL-8 and TNF-α

3.3.1. Assay principle

Human normal keratinocytes (HNK, isolated from biopsy obtained from surgical operation) are plated in 96-well microplate at the density of 20 000 cells/well (culture medium KSFM – Gibco® Invitrogen). Plate is placed in an incubator maintained at 37°C and 5% CO₂ until cells are confluent at 80%.

3.3.2. Reagents

- Interleukin-1β: recombinant human IL1β, 201-LB, R&D systems.
- Phorbol Myristate Acetate (PMA): Sigma P-8139.
- LBG 10602 tested at 2.5- 1.5- 1 and 0.5 %
- Dexametazone: Sigma, D-4902.



3.3.3. Cell treatments

Cells are pre-treated at different concentrations of LBG 10-602 in culture medium for 24H and then stimulated with IL1 β (1 ng/ml for IL8 release) or Phorbol Myristate Acetate at 500 ng/ml (for TNF α) during 24H. Each product concentration is evaluated on 3 wells. Cells are kept without treatment for control condition.

At the end of the experiment, cell supernatants are kept at -20 °C. Determination of protein is performed in each well.

3.3.4. Cytotoxicity assay

After 48h incubation period, cellular wells are rinsed with PBS without Ca⁺⁺/Mg⁺⁺ (Gibco® Invitrogen). XTT mixture is added to each well and the plate incubated for 3h.

The spectrophotometrical absorbance is measured using a microtiter plate (Safire, Tecan). The wavelength to measure absorbance of the formazan is 450 nm.

Viability from cells treated with the products are defined as percentage compared with control 100% (non treated cells):

$$\% = [(OD_{\text{Treatment}} / OD_{\text{control}})] \times 100$$

Absorbance measurement from 4 wells treated with the same product concentration are averaged. This average is compared with control average (4 wells also) : Student's t test – significant difference at 95% if $p < 0.05^*$ and at 99% if $p < 0.01^{**}$.

3.3.5. Cytokines assays

Cytokines are measured as described in the Quantikine® human IL-8 (R&D Systems, D8000C), Quantikine® human TNF- α (R&D Systems, DTA00C)

The spectrophotometrical absorbance is measured using a scanning multiwell spectrophotometer (Safire, Tecan). The wavelength to measure absorbance of the substrate is 450 nm.

A best fit curve is generated for the standards in order to extrapolate relative samples values. The amount of IL-8 and TNF- α are expressed in pg/ml.

A protein assay is purchased for every well of the plate.

The percentages of inhibition induced by the treatment are calculated for each concentration of LBG10-602.

$$\% \text{ Inhibition} = 100 \times [1 - (\text{stimulated and treated value} / \text{not stimulated control value})]$$

$$\% \text{ Inhibition of the stimulation} = 100 \times [1 - (\text{stimulated and treated value} / \text{control stimulated value})]$$

3.3.6. Protein synthesis

Principle of this assay is the bicinchoninic acid (BCA) assay which is similar to the Lowry procedure. Determination of protein is performed in each well (cell lysate) for which supernatant was quantified for cytokines. Values are expressed in μg protein.

Therefore, final results for IL8 or TNF- α are expressed in pg/ml / μg protein.

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Values from 3 wells treated with the same condition are averaged. This average is compared with control average: Student's t test – significant difference at 95% if $p < 0.05^*$ and at 99% if $p < 0.01^{**}$.

3.4. Results

3.4.1. Cell cytotoxicity

KHN treatment with LBG10-602 for 48H

	Control	2.5%	2%	1.5%	1%	0.5%	0.25%
Absorbance 450 nm	1.6097 ± 0.0915	1.6471 ± 0.0783	1.6848 ± 0.0414	1.5209 ± 0.0851	1.6618 ± 0.1405	1.7044 ± 0.0943	1.5371 ± 0.0207
% viability	100	105	108	97	106	109	98
p (Student)		0.1707	0.0870	0.3050	0.1787	0.0729	0.3522

3.4.2. Interleukine-8 quantification

	IL-8 (pg/ml/μg proteins)	% Inhibition	IL-8 (pg/ml/μg proteins)	% Inhibition
Control	21 ± 5,6		71.3 ± 9	
LBG10-602 2.5%	8.7 ± 0.7	58% **	23.3 ± 3.4	67% **
LBG10-602 1.5%	14.6 ± 0,5	30% ns	38.9 ± 7.9	45% **
LBG10-602 1%	11.6 ± 0,6	45% *	40.3 ± 0.6	43% **
LBG10-602 0.25%	15.5 ± 0,1	26% ns	44.3 ± 12.5	38% *
X	ND		30.5 ± 4.9	57% **

Means of 3 wells ± SD

** : $p < 0.01$ (Student's t test) – significant results at 99%.

* : $p < 0.05$ (Student's t test) – significant results at 95%.

ns: not significant

3.4.3. TNF-α quantification

	TNF-α (pg/ml/μg proteins)	% Inhibition	TNF-α (pg/ml/μg proteins)	% Inhibition
Control	not detectable		7.3 ± 0.5	
LBG10-602 2.5%	non detectable		4.6 ± 0.7	37% **
LBG10-602 1.5%	0,3		4.5 ± 0.4	39% **
LBG10-602 1%	0,0 ± 0.0		5.1 ± 0.2	30% **
LBG10-602 0.25%	0,2 ± 0.0		6.1 ± 1.2	15% ns
X	ND		3 ± 0.7	59% **

Means of 3 wells ± SD

** : $p < 0.01$ (Student's t test) – significant results at 99%.

* : $p < 0,05$ (Student's t test) –significant results at 95%.

ns: not significant



3.5. Conclusion

- In normal human keratinocytes (NHK), LBG 10602 is not cytotoxic at **0.25 % to 2.5 %**.
- **LBG 10602 inhibits IL8 release upon IL1 β cell stimulation or not at 2.5%** (58 % inhibition and 67 %, respectively for not stimulated and stimulated cells) and at 1 % (45 % inhibition and 43 %, respectively for not stimulated or stimulated cells. These results are very significant, $p < 0.01$.
- **LBG 10602 inhibits TNF- α release at 2.5 %, 1.5 % and 1%.** (% inhibition 37%, 39 and 30%, respectively $p < 0.01$. Dexametasone used as positive control inhibits the production of this mediator by 59 % at 10 μ M.

In conclusion, LBG 10602 is a good anti-inflammatory compound.

3.6. References

Baggiolini M and Clark-Lewis I: interleukin 8, a chemotactic and inflammatory cytokine. FEBS Lett. 385 : 97-1101, 1992.

4. *In vitro* assay of anti-vasodilation properties : measure of PGE₂

Effect of LBG10-602 on PEG₂ release in normal human keratinocytes.

4.1. Introduction

The mediators arising from the activation of phospholipase A2, in particular Prostaglandin E₂ (PGE₂) play a pivotal role in cutaneous inflammatory reactions (1).

In the skin, PGE₂ is generally implicated in disorders linked to a dysregulation of inflammatory reactions; for example, solar erythema (2), eczema, contact dermatitis, (3) or psoriasis. Human Cathelicidin CCL-37 stimulates the production of PGE₂ in human keratinocytes. Furthermore, a COX-2 inhibitor inhibits the anti-apoptotic effect of LL37 in psoriasis (4, 5, 6). PGE₂ is also implicated in cutaneous hyperreactivity and pain sensations.

Normal human keratinocytes (KHN) release PGE₂ upon stimulation with Phorbol Myristate Acetate (PMA).

The aim of this study is to evaluate the effects of LBG 10-602 treatment on PGE₂ release in NHK.

4.2. Cytotoxicity

In a previous study, XTT test showed that LBG 10-602 did not present any cytotoxicity effects after 48H of treatment with LBG 10-602 at 1.5 %, 1%, 0.25 % and 0.1 %.



4.3. Effect of LBG10-602 on Prostaglandin E₂ release

4.3.1. Test

Human normal keratinocytes (HNK, isolated from biopsy obtained from surgical operation) are plated in 96-well microplate at the density of 20 000 cells/well (culture medium KSFM – Gibco® Invitrogen). Plate is placed in an incubator maintained at 37°C and 5% CO₂ until cells are confluent at 80%.

Cells are treated at different concentrations in culture medium for 24H with PMA. Each product concentration is evaluated on 3 wells. Cells are kept without treatment for control condition. Cell supernatant are kept at -20°C until the measurement. Quantification of PGE₂ is performed using an ELISA kit (Enzo® Life sciences ADI-900-001).

4.3.2. Reagents

LBG10-602: sterile stock solution diluted in culture medium and supplied at 4°C.

Tested concentrations: 1.5% - 1% - 0.25% and 0.1% (v/v).

Indomethacin: control inhibitor of PGE₂ release used at 1 µM (Sigma I-7378).

Phorbol Myristate Acetate: PMA (Sigma P-8139). Cell stimulation at 1ng/ml.

4.3.3. PEG₂ release

After 24H of treatment, the production of PGE₂ in cell supernatant is measured using an ELISA method described in the EIA kit Prostaglandin E₂ – Enzo® Life sciences - ADI-900-001, lot 08241002 (405 nm).

4.3.4. Protein synthesis

Principle of this assay is the bicinchoninic acid (BCA) assay which is similar to the Lowry procedure. Determination of protein is performed in each well (cell lysate). Values are expressed in µg protein.

4.3.5. Expression of the results

Therefore, final results for PGE₂ quantification are expressed in pg/ml/µg protein.

Values corresponding to 3 wells treated with the same condition are averaged. This average is compared with control average: Student's t test – significant difference at 95% if $p < 0.05^*$ and at 99% if $p < 0.01^{**}$.

The percentages of inhibition are calculated for each concentration of LBG10-602 with or without stimulation.

% Inhibition = $100 \times [1 - (\text{not stimulated treated value} / \text{not stimulated control value})]$

% Inhibition of stimulation = $100 \times [1 - (\text{stimulated treated value} / \text{stimulated value control})]$

4.4. Results

Effect of LBG 10-602 on PGE₂ release induced by PMA:



	PGE ₂ (pg/ml/μg proteins)	% Inhibition	PGE ₂ (pg/ml/μg proteines)	% Inhibition
Control	180.7 ± 81.4		442.3 ± 122.5	
LBG10-602 1%	83.5 ± 10,2	54%	165.5 ± 43.9	63% *
LBG10-602 0,25 %	102.7 ± 30,8	43% ns	162.7 ± 53.3	63% *
LBG10-602 0,1%	107.4 ± 37.7	41% ns	160.9 ± 8.2	64% **
X	no determinated		49.4 ± 9.4	89% **

Means of 3 wells ± SD

** : p < 0,01 (Student's t test) – significant results at 99%.

* : p < 0,05 (Student's t test) – significant results at 95%. ns: not significant

4.5. Conclusion

The treatment of Human Keratinocytes with LBG10-602 at 1% ; 0.25% and 0.1% inhibits the production of PGE₂ induced by PMA. This result is statistically significant (% inhibition 63 %, 63 % and 64 %, respectively).

In conclusion, the inhibitory effect of LBG-10601 on PGE₂ release by NHK stimulation is of particular interest and suggests that this product could be useful in controlling inflammatory skin disorders.

5. *In vitro* assay on VEGF release inhibition

Effect on LBG10-602 on VEGF release by Normal Human Keratinocytes

5.1. Introduction

Vascular endothelial growth factor (VEGF) is a major angiogenic factor that stimulates vasculogenesis and angiogenesis. Its normal function is to create new blood vessels during embryonic development, new blood vessels after injury, muscle following exercise, and new vessels to bypass blocked vessels. In homeostatic conditions, VEGF is necessary for wound healing and to maintain a healthy skin.

VEGF is produced by fibroblasts in dermis but also by keratinocytes. *In vivo*, cytokines like IL1 or TNF-α modulate the activity of VEGF.

Normal human keratinocytes express and release VEGF upon the stimulation with IL1α.

The aim of this study is to test the effect of LBG 10-602 on VEGF release by NHK.

5.2. Cytotoxicity – XTT assay

The aim of this study is to evaluate the cytotoxicity of LBG 10-602 on normal human keratinocytes (primocultures).

XTT assay (non radioactive and colorimetric method allows to quantify viability and proliferation of cells through their mitochondrial deshydrogenase activity. This is an easy and accurate method, with reproducible results.

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5.2.1. Assay principle

The assay is based on the cleavage of the yellow tetrazolium salt XTT to form an orange formazan dye by metabolic active cells. Therefore, this conversion only occurs in viable cells. The formazan dye formed is soluble in aqueous solutions and is directly quantified using a scanning multiwell spectrophotometer (ELISA reader, wavelength 450nm).

5.2.2. Reagents

"Cell Proliferation Kit II (XTT)" (Roche) contains XTT labeling reagent and PMS (N-methyl dibenzopyrazine methyl sulfate, electron-coupling reagent). The mixture is prepared freshly at 0.3 mg/ml final concentration in culture medium KSFM (Gibco Invitrogen).

5.2.3. Product to be tested

LBG10-602 was supplied in liquide form: 2 bottles of 2 ml sterile stock solution to be diluted in culture medium. Keeping at 4°C.

Tested concentrations: 2.5 - 2 - 1.5 - 1 - 0.5% and 0.25% (v/v, dilutions in medium culture).

5.2.4. Cell culture and treatment

Human normal keratinocytes (HNK, isolated from biopsy obtained from surgical operation) are plated in 96-well microplate at the density of 20 000 cells/well (culture medium KSFM – Gibco® Invitrogen). Plate is placed in an incubator maintained at 37°C and 5% CO₂ until cells are confluent at 80%.

Cells are treated at different concentrations in culture medium for 48h. Each product concentration is evaluated on 4 wells. Cells are kept without treatment for control condition.

5.2.5. Cytotoxicity measurement

After 48h incubation period, cellular wells are rinsed with PBS without Ca⁺⁺/Mg⁺⁺ (Gibco® Invitrogen). XTT mixture is added to each well and the plate incubated for 3h. The spectrophotometrical absorbance is measured using a microtiter plate (Safire, Tecan). The wavelength to measure absorbance of the formazan is 450 nm.

Absorbance measurement from 4 wells treated with the same product concentration is averaged. This average is compared with control average (4 wells also): Student's t test – significant difference at 95% if p<0.05* and at 99% if p<0.01**.

Viability from cells treated with the products is defined as percentage compared with control 100% (no treated cells):

$$\% = [(OD_{\text{Treatment}} / OD_{\text{control}})] \times 100$$



5.3. Quantification of VEGF

5.3.1. Assay principle

Human normal keratinocytes (HNK, isolated from biopsy obtained from surgical operation) are plated in 96-well microplate at the density of 20 000 cells/well (culture medium KSFM – Gibco® Invitrogen). Plate is placed in an incubator maintained at 37°C and 5% CO₂ until cells are confluent at 80%.

5.3.2. Reagents

Interleukin-1 β : recombinant human IL1 β , 201-LB, R&D systems. Cell stimulation at 10 ng/ml.
LBG 10-602: tested at 2.5 %-1.5%-1% and 0.25 %
Dexametazone: Sigma, D-4902 tested at 10 μ M.

5.3.3. Cell treatments

Cells are pre-treated at different concentrations of LBG 10602 in culture medium for 24H and then stimulated with IL1 β (10 ng/ml) for 24H. Each product concentration is evaluated on 3 wells stimulated or not. Cells are kept without treatment for control condition.
At the end of the experiment, cell supernatants are kept at -20 °C.
Determination of protein is performed in each well.

5.3.4. VEGF assay

VEGF is measured as described in the VEGF ELISA kit (Thermo Scientific, EH VEGF).

The spectrophotometrical absorbance is measured using a scanning multiwell spectrophotometer (Safire, Tecan). The wavelength to measure absorbance of the substrate is 450 nM.

A best fit curve is generated for the standards in order to extrapolate relative samples values. The amount of VEGF is expressed in pg/ml. A protein assay is purchased for every well of the plate.

The percentages of inhibition induced by the treatment are calculated for each concentration of LBG10-602 in the presence or not of the stimulation.

% Inhibition = $100 \times [1 - (\text{stimulated and treated value} / \text{not stimulated control value})]$

% Inhibition of the stimulation = $100 \times [1 - (\text{stimulated and treated value} / \text{control stimulated value})]$

5.3.5. Protein synthesis

Principle of this assay is the bicinchoninic acid (BCA) assay which is similar to the Lowry procedure. Determination of protein is performed in each well (cell lysate) for which supernatant was quantified for cytokines. Values are expressed in μ g protein.

Therefore, final results for VEGF quantification are expressed in pg/ml / μ g protein.

Values from 3 wells treated with the same condition are averaged. This average is compared with control average: Student's t test – significant difference at 95% if $p < 0.05^*$ and at 99% if $p < 0.01^{**}$.



5.4. Results

5.4.1. Cytotoxicity: % viability

Treatment of KHN with LBG10-602 during 48H.

	Control	2.5%	2%	1.5%	1%	0.5%	0.25%
Absorbance 450 nm	1.6097 ± 0.0915	1.6471 ± 0.0783	1.6848 ± 0.0414	1.5209 ± 0.0851	1.6618 ± 0.1405	1.7044 ± 0.0983	1.5371 ± 0.0207
% viability	100	105	108	97	106	109	98
p (Student)		0.1707	0.0870	0.3050	0.1787	0.0729	0.3522

5.4.2. VEGF release

	VEGF (pg/ml/μg proteins)	% Inhibition	VEGF (pg/ml/μg proteins)	% Inhibition
Control	16.4 ± 2,8		18.3 ± 2,6	
LBG10-602 2.5%	8.8 ± 2,9	46% *	11.5 ± 0,8	37% *
LBG10-602 à 1.5%	12 ± 0,6	27% *	12.3 ± 1,1	33% *
LBG10-602 à 1%	13.1 ± 0,6	20% ns	14.1 ± 3,5	23% ns
LBG10-602 à 0.25%	16.5 ± 3,6		10.9 ± 2,7	40% *
X	non déterminé		8.1 ± 1,5	56% **

Means of 3 values ± SD

** : p < 0.01 (Student's t test) – significant results at 99%.

* : p < 0.05 (Student's t test) – significant results at 95%.

ns: not significant

5.5. Conclusion

LBG 10602 is not cytotoxic in human keratinocytes at 0.25 % to 2.5 %.

NHK treatment with LBG 10-602 at 2.5 % and 1.5 % inhibits the production of VEGF induced by IL1β (inhibition 37%, p < 0.05 and 33 %, p < 0.01, respectively)

This product presents an interest to improve vascular disorders linked to cutaneous redness or blotches for exemple.



6. *In vitro* evaluation on neo vessels development: measure of neo vessels creation

6.1. Angiogenesis test

Skin endothelial cells HskMEC tend to grow and associate themselves to create new vessels when they are cultured in a three dimensional support, such as Matrigel™. Incubation of these cells with different compounds, enables to see if such products have pro or anti-angiogenesis properties.

6.2. Materials and methods

Angiogenesis was performed on 96-well plates coated with Matrigel™ (BD biosciences, San José, CA). Skin endothelial cells HskMEC were briefly treated with trypsin, washed and were seeded at $1.5 \cdot 10^4$ cells/cm². Cultures were maintained in normoxic (20% O₂) conditions.

Cell rearrangement and pseudo-vessels formation were visualized using a Zeiss Axiovert 200M fluorescence inverted microscope (Zeiss, Le Pecq, France) equipped with an Axiocam high resolution numeric camera linked to a computer driving the acquisition software Axiovision (Zeiss). The direct real-time visualisation of the formation of pseudo-vessels was monitored during 24h. Angiogenesis was quantified by the determination of pseudovessel number, longer than 50 micrometers using Axiovision software.

6.3. Results

Cell cytotoxicity of product LBG-10-602 was evaluated using the Alamar Blue test. It was determined that product LBG-10-602 is not cytotoxic for concentrations lower than 2,5%. At 2,5%, the product is not cytotoxic during 24 hours, and shows a slight cytostatic effect at t=72 hours.

Three independent experiments were performed with the product LBG-10-602 at two concentrations: 1,25% and 2,5%.

Experiment 1

Concentration in %	0	1,25	2,5
Nb of pseudovessels	50	ND	18

Experiment 2

Concentration in %	0	1.25	2.5
Nb of pseudovessels	51	9	0

Experiment 3

Concentration in %	0	1.25	2.5
Nb of pseudovessels	70	0	0



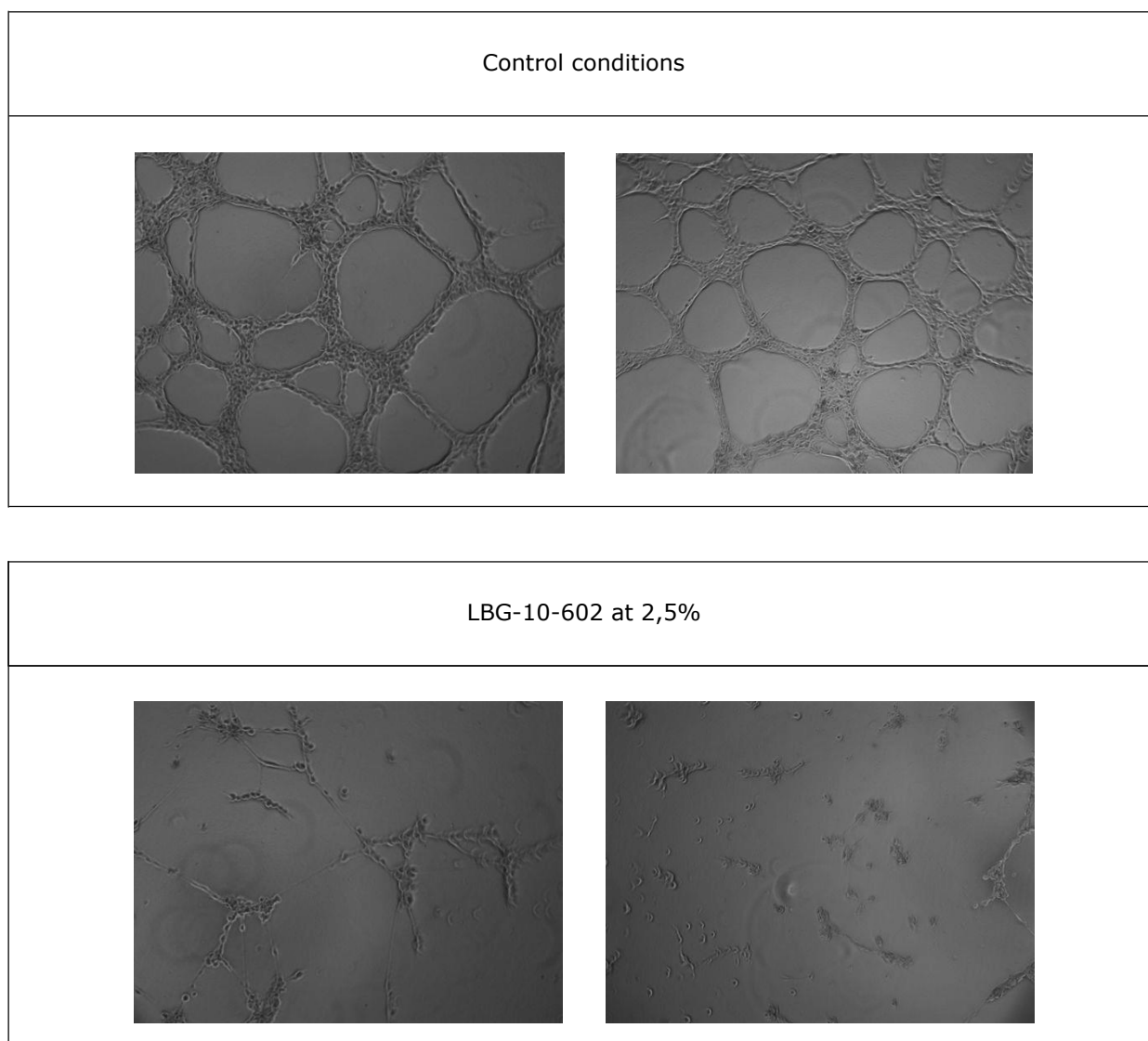
6.4. Conclusions

Product LBG-10-602 shows a clear inhibitory effect on the formation of neovessels by skin endothelial HskMEC, suggesting that this product could hinder the formation of new blood vessels in an irritated skin.

This work has been performed by the team of Dr Kieda at CNRS CBM Orléans by Mr Mahdi Nadim and Dr Catherine Grillon, with the helpful support of the cytometry platform of CRNS /CBBM at Orléans.

6.5. Annex

Pictures of the cells in the Matrigel™





7. Clinical evaluation versus placebo under dermatological control

7.1. Aim of the study

This study aims at assessing the efficacy of a serum vs. placebo on 20 women suffering from rosacea. The various biometric measurements as well as pictures were taken during the first visit and after 28 days of bi-daily application.

7.2. Experimental design

7.2.1. Design of the study

The activity of the serum was assessed by 20 women according to the inclusion criteria defined with the promoter. The measurements were taken in a controlled-atmosphere room ($22^{\circ}\text{C} \pm 2^{\circ}\text{C}$), after stabilisation of the volunteers for at least 10 minutes.

The various biometric measurements as well as pictures were taken on both the cheekbones. The volunteers used the serum and placebo on half their face, every day, twice daily from D0 to D28. Each volunteer was supposed to notify laboratoire BIO-EC of any discomfort or undesirable event that would occur. They did not stop or change the frequency of application without prior notice. During the D28 visit, the biometrological measurements were performed in the same way as on D0.

7.2.2. Sequence of operations for each volunteer

Recruitment: Week 02 (Caucasian women with rosacea).

First visit on D0: (Week 03), the technician and the dermatologist in charge of the study,

- Have the volunteers stabilise in a controlled-atmosphere room for 10 minutes
- Check the inclusion and exclusion criteria,
- Notes down the potentially associated treatments
- Have the volunteer sign the consent form,
- Take the colour measurement and picture on the cheekbone
- Distributes the product and the questionnaire

From D0 to D28, the volunteers applied the serum morning and evening on half their face according to the recommendations.

Second visit on D28: (Week 07), the technician and the dermatologist in charge of the study,

- Have the volunteers stabilise in a controlled-atmosphere room for 10 minutes
- Record the undesirable events that occurred during use of the product
- Take the colour measurement and picture on the cheekbone
- Have the volunteer sign the compensation sheet.

7.3. Volunteers

7.3.1. Inclusion and exclusion criteria

Inclusion criteria

- Caucasian women with rosacea on the cheekbone, aged over 30, that have signed the consent form
- Agreeing on not using face products for the length of the study, other than those provided
- Not having applied any cosmetic product since the eve of the appointment.



Exclusion criteria

- Pregnant or nursing women
- Subjects with dermatological problems and/or a known allergy to cosmetic products and/or to one of the cream constituents.
- Subjects under medical and/or dermatological treatment likely to affect the measurements.
- Subjects having participated in a clinical or cosmetic trial in the 7 days prior to D0, or participating in a clinical or cosmetic trial.

7.3.2. Volunteers included

Altogether, 20 women matching the inclusion and exclusion criteria defined in the protocol were included in the study. They were informed of the possible drawbacks linked with the application of the product and the technical conditions in which the assessment is performed. They freely signed the consent form which was written in compliance with the Declaration of Helsinki and the December 20th, 1988 act of the Code de la Santé Publique.

Demographic characteristics

The demographic characteristics of the volunteer group (mean $\pm$ SD) are the following:

Serum anti-rougeur	N = 20 women
	Ages : 53 $\pm$ 12 years

Schedule compliance

No volunteer left the study prematurely.

Concomitant treatment

The volunteers included in the study did not take any concomitant treatment likely to induce a modification of the cutaneous state.

7.4. Products

7.4.1. Products studied

The serums to test were identified as follows: Serum anti-rougeur Ref : 001PL1.A and Serum anti-rougeur Ref : 001PL1.B

It was a whitish serum packaged in a glass pump dispenser. The products tested were packaged by the promoter.

7.4.2. Use instructions for the product

The serums were given to the volunteers on D0 and used on half their face twice daily until D27 in the evening.

The amount of serum to apply was left to the volunteers' discretion, according to their habits. The possible remarks were noted in the final interview.



7.5. Methods

7.5.1. Dermatological evaluation

The dermatologist scored both sides of the face before and after treatment using the following scale:

0 1 2 3 4 5
|-----|-----|-----|-----|-----|

7.5.2. Measurement of the colour red with the spectroradiometer

Colour measurements were made according to the following parameters:

- Standard field of vision: 2° as defined by CIE in 1931
- Illuminant D65: corresponding to daylight
- Area of measure: 3 mm
- Colour spaces used: L^* a^* b^* et L^* C^* h^*
- SCI method (specular included)

Study of the a^* axis:

Using the spectroradiometer, the parameter a^* will be followed, which corresponds to the red axis of the $L^*a^*b^*$ referential. If this value decreases, the area observed is less red.

7.5.3. Measurement of temperature

The cutaneous temperature is measured using a very small sensor allowing for a shorter response time and enabling to be freed from conduction problems (thermal memory phenomenon) often encountered on this type of measurement. It is expressed in Celsius degrees and is generally around 30°C; it mainly varies with the exterior temperature, for instance the temperature of the skin can reach 35°C when the outside temperature is 34°C.

The temperature of the skin was measured on the rosacea areas.

7.5.4. Analysis of colour using the Siascope

Siascopy understands the way that light interacts with the skin; the manner in which it scatters or bounces, the amount absorbed by cells and other structures along with the different changes in wavelength or colour. By understanding these interactions and comparing the light sent into skin with light that comes back out, Siascopy is able to determine the nature and position of many of the different components of the skin. Siascopy measures key chromophores within the skin, in particular haemoglobin, melanin and collagen.

When the scanner illuminates the skin, some of the light is reflected and scattered from the surface. The remainder is transmitted into the top layers of the skin.

Varying fractions of the incoming light are then firstly absorbed by the melanin in the epidermis before entering the dermis, where they are absorbed by the haemoglobin in the blood vessels. Scattering also occurs in the dermis when the light interacts with the collagen, resulting in a portion of the light being remitted back to the surface.

By interpreting the combination of wavelengths that are received back by the Siascope, Siascopy is then able to produce Siascans; these are generated by referring to inbuilt proprietary mathematical models of skin optics.



7.6. Results

7.6.1. Atmospheric conditions around Paris

Maximal and minimal temperatures around Paris during the study were:

- January 2008: -3.6°C to 9.5°C
- February 2009: -3.9°C to 17.4°C

7.6.2. Statistical method

The basic statistical parameters (mean and standard deviation) were calculated for each timepoint and parameter measured. Then, the assessment of the overall effect of the product studied was made by calculating the variation percentage compared to the initial measurement.

In order to determine whether the evolution noticed is significant or not, a Student's t-test was performed. The statistical analysis was made with Student's t-test on paired samples. The assumptions are the randomness and normal distribution of the samples.

7.6.3. Protocol deviation

All inclusion and evaluation criterion were respected.

7.6.4. Undesirable events

No undesirable event occurred during the study.

7.6.5. Dermatological evaluation

No intolerance reaction was observed by the dermatologist

The dermatologist concluded to a very good tolerance of the products tested.

7.6.6. Results on the colour red using the Siascope

The mean values of the measures of haemoglobin using the Siascope are presented in the tables below.

Results on 50 % of the subjects:

Volunteer	Active				Placebo			
	T0	T28	Difference	% variation	T0	T28	Difference	% variation
4	216.380	199.387	-16.993	-8%	204.281	183.257	-21.024	-10%
5	241.152	215.152	-26.000	-11%	216.204	209.515	-6.689	-3%
6	241.617	257.647	16.030	7%	240.847	258.661	17.814	7%
7	203.721	198.016	-5.705	-3%	215.980	208.095	-7.885	-4%
8	188.168	193.096	4.928	3%	179.555	209.811	30.256	17%
10	157.742	140.642	-17.100	-11%	158.123	146.286	-11.837	-7%
12	184.331	158.198	-26.133	-14%	161.679	146.736	-14.943	-9%
13	176.260	188.901	12.641	7%	178.173	209.023	30.850	17%
18	216.423	227.881	11.458	5%	195.804	215.858	20.054	10%
19	236.754	185.515	-51.239	-22%	186.880	193.857	6.977	4%
Mean	206.255	196.444	-9.811	-5%	193.753	198.110	4.357	2%
SD	29.196	33.098	21.623		26.1	33.4	19.3	



After 28 days of application, an improvement of rosacea is observed when comparing the active to the placebo (improvement of 7 %).

Results on 30 % of the subjects:

Volunteer	Active				Placebo			
	T0	T28	Difference	% variation	T0	T28	Difference	% variation
5	241.152	215.152	-26.000	-11%	216.204	209.515	-6.689	-3%
8	188.168	193.096	4.928	3%	179.555	209.811	30.256	17%
12	184.331	158.198	-26.133	-14%	161.679	146.736	-14.943	-9%
13	176.260	188.901	12.641	7%	178.173	209.023	30.850	17%
18	216.423	227.881	11.458	5%	195.804	215.858	20.054	10%
19	236.754	185.515	-51.239	-22%	186.880	193.857	6.977	4%
Mean	207.181	194.791	-12.391	-6%	186.383	197.467	11.084	6%
SD	28.120	24.381	25.994		18.5	25.9	19.2	

After 28 days of application, an improvement of rosacea is observed when comparing the active to the placebo (improvement of 12 %).

7.6.7. Results on melanin using the Siascope

The mean values of the measures of haemoglobin using the Siascope are presented in the table below:

	Silidine A				Placebo			
	T0	T28	Difference	% variation	T0	T28	Difference	% variation
Mean	241.89	239.04	-2.86	-1%	224.00	239.88	15.88	7%
SD	41.94	38.37	16.30		40.99	35.18	19.83	

After 28 days of application, a decrease of melanin is observed when comparing the active to the placebo (improvement of 8 %).

7.7. General conclusions

The aim of this study was to assess over 28 days on a volunteer panel suffering from rosacea the efficacy of a serum vs. placebo:

- **Dermatologist:**

The serum was tested under dermatological control and no adverse cutaneous reaction was observed.

The dermatologist concluded to a very good tolerance of the products tested.

- **Results on rosacea :**

After 28 days of application, a 12% improvement of rosacea is observed on the cheekbones.



- **Results on melanin:**

After 28 days of application, a 8% decrease of melanin is observed on the cheekbones.

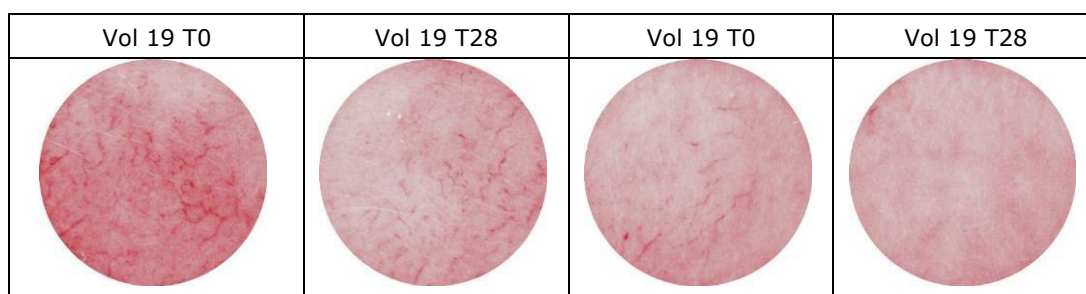
7.8. Appendixes

Illustration of the evolution of rosacea

Active		Placebo	
Vol 04 T0	Vol 04 T28	Vol 04 T0	Vol 04 T28
Vol 05 T0	Vol 05 T28	Vol 05 T0	Vol 05 T28
Vol 06 T0	Vol 06 T28	Vol 06 T0	Vol 06 T28
Vol 07 T0	Vol 07 T28	Vol 07 T0	Vol 07 T28



Active		placebo	
Vol 08 T0	Vol 08 T28	Vol 08 T0	Vol 08 T28
Vol 10 T0	Vol 10 T28	Vol 10 T0	Vol 10 T28
Vol 12 T0	Vol 12 T28	Vol 12 T0	Vol 12 T28
Vol 13 T0	Vol 13 T28	Vol 13 T0	Vol 13 T28
Vol 18 T0	Vol 18 T28	Vol 18 T0	Vol 18 T28



Results on Melanin

Vo-lontaire	Actif				Placebo			
	T0	T28	Différence	% varia-tion	T0	T28	Différence	% variation
1	236.423	261.250	24.827	11%	209.686	251.745	42.059	20%
2	179.400	176.974	-2.426	-1%	181.254	194.638	13.384	7%
3	171.392	166.868	-4.524	-3%	159.847	187.476	27.629	17%
4	250.384	253.324	2.940	1%	233.036	254.524	21.488	9%
5	179.755	191.865	12.110	7%	165.656	186.166	20.510	12%
6	217.296	208.510	-8.786	-4%	172.735	206.606	33.871	20%
7	221.802	240.813	19.011	9%	211.149	221.597	10.448	5%
8	251.532	248.927	-2.605	-1%	232.770	251.316	18.546	8%
9	280.600	271.446	-9.154	-3%				
10	293.015	264.837	-28.178	-10%	284.322	274.736	-9.586	-3%
11	215.101	197.524	-17.577	-8%	216.627	217.118	0.491	0%
12	297.046	287.052	-9.994	-3%	295.493	303.081	7.588	3%
13	263.723	254.731	-8.992	-3%	263.418	246.345	-17.073	-6%
14	230.702	247.128	16.426	7%	205.466	237.987	32.521	16%
15	302.603	310.662	8.059	3%	220.699	283.780	63.081	29%
16	275.787	265.746	-10.041	-4%	258.953	279.290	20.337	8%
17	275.129	266.807	-8.322	-3%	251.578	236.257	-15.321	-6%
18	268.302	226.176	-42.126	-16%	261.322	260.997	-0.325	0%
19	251.399	246.453	-4.946	-2%	257.936	269.657	11.721	5%
20	176.452	193.627	17.175	10%	174.068	194.331	20.263	12%
Moyenne	241.892	239.036	-2.856	-1%	224.001	239.876	15.875	7%
Ecart type	41.942	38.368	16.298		41.0	35.2	19.8	